HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented automated assembly line testing procedures using **Python scripts**, reducing manual test time by **30%** for a key product family.
- Developed and maintained **Java Spring Boot** REST APIs to integrate manufacturing execution systems with enterprise resource planning software, improving data synchronisation accuracy by **15%**.
- Authored comprehensive **Standard Operating Procedures (SOPs)** and work instructions for new automation equipment, ensuring clear documentation for production teams.
- Collaborated with external **machine builders and system integrators** to define technical requirements and conduct Factory Acceptance Tests (FAT) for robotic assembly cells.
- Optimised **SQL queries** against a PostgreSQL database to extract production yield data, supporting the generation of weekly operational performance reports.
- Contributed to the adoption of **Lean Manufacturing principles**, specifically participating in 5S initiatives to improve workshop organisation and efficiency.
- Containerised key manufacturing support services using **Docker** and deployed them to an **AWS EC2** environment for improved reliability and scalability.

SKILLS

Manufacturing Technologies – Automation, Process Improvement, Equipment Modernisation, Lean Manufacturing, 5S, Continuous Improvement, Production Processes, Industrial Technologies

Software Development – Java, Spring Boot, Python, SQL, REST APIs, Docker, Git, Jenkins

Cloud & Infrastructure – AWS (EC2, S3, CloudWatch), Linux, Shell Scripting

Data & Analytics – Power BI, Tableau, DAX, Power Query, Data Modelling, Data Warehousing, pandas, NumPy, scikit-learn

Documentation & Compliance – SOPs, URS, RFQs, Technical Documentation, Bills of Materials (BOMs), Occupational Health and Safety, International Trade Compliance (ITC)

Project Management – Agile/Scrum, Requirements Gathering, UAT Coordination, Change Management, Investment Proposals

PROJECTS

Manufacturing Automation Integration Platform

- Developed a **Python-based automation framework** to orchestrate data flow between disparate manufacturing equipment and a central logging service, reducing manual data entry errors by **22%**.
- Built **REST APIs using Flask** to enable real-time status updates from PLC controllers to a cloud-based monitoring dashboard, improving visibility into production line status.
- Created automated test scripts to validate integration points, ensuring data integrity and timely processing across the manufacturing technology stack.

Production Data Analytics Dashboard

- Designed and implemented an interactive **Power BI dashboard** to visualise key production metrics including yield, cycle time, and equipment uptime, enabling faster operational decision-making.
- Developed **DAX measures** and Power Query scripts to transform and integrate data from SQL databases and flat files into a cohesive data model for reporting.
- Presented the final dashboard to manufacturing engineering and operations teams, incorporating feedback to refine visualisations and reports.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)